TWINFLOW AI DIGITAL TWIN

Intelligent Spatio-Temporal Warehouse Logistics System

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Executive Audit Q4

Scope: Comprehensive congestion audit, AGV flow telemetry, and DSTGT spatial prediction benchmark for **Weekly Congestion**.

Active AGV Fleet	Peak Congestion Risk	Predicted Queue Lead	Model Confidence
18 / 22	**87%**	**22 AGVs**	**92.4%**
81.8% Operational	High Risk (Aisle B)	Horizon: +15 mins	DSTGT Spatio-Temporal

1. Warehouse Zone & Hotspot Risk Breakdown

Zone / Location	Current Status	Occupancy Rate	Predicted Queue	Delay Impact	AI Risk Level
Aisle B (Storage Hotspot) Racks B1-B8	**CRITICAL**	**88% (53 / 60)**	22 AGVs queued	**+6.4 min delay**	**87% (High Risk)**
Junction J-04 (Sortation Hub) Zone 3 Main Feeder	**WARNING**	**82% (41 / 50)**	14 AGVs queued	**+4.1 min delay**	**76% (Moderate)**
Aisle C (Buffer Storage) Zone 2 Bypass	**OPTIMAL**	58% (32 / 55)	3 AGVs queued	**+0.8 min delay**	**24% (Low Risk)**
Inbound Dock 2 Zone 1 Receiving	**RESOLVED**	62% (28 / 45)	5 AGVs staged	**+1.4 min delay**	**31% (Low Risk)**

2. DSTGT Model Attention Feature Attributions

Attention weights explain why the Spatio-Temporal Graph Transformer flagged Aisle B as a primary bottleneck:

Feature Parameter	Attention Weight	Real-Time Metric Value	Graph Impact Analysis
Aisle B Physical Density	**0.32 (32%)**	88% rack fill rate	Direct node saturation across topological edge path
AGV Fleet Inflow Velocity	**0.24 (24%)**	1.2 m/s head-to-tail	Consecutive robot arrivals exceeding dock discharge bandwidth
Temporal Queue Growth	0.18 (18%)	22 robots queued	Temporal attention detects +6.4m queue build over 15m window
Order Wave Batch Volume	0.15 (15%)	+35% surge order batch	E-commerce priority wave scheduled simultaneously
Conveyor Motorized Load	0.11 (11%)	92% capacity load	Backpressure on sortation feeder junction J-04

3. Prescriptive Rerouting & Optimization Recommendations:

• **Dynamic Rerouting:** Divert 4 inbound AGVs carrying standard non-priority SKUs from Aisle B to Aisle C (Buffer Storage) to reduce risk by 22%.
• **Speed Adjustment:** Reduce downstream sortation line speed by 10% to prevent physical pallet stacking at Junction J-04.
• **Wave Staggering:** Stagger subsequent inbound batch wave by 12 minutes until Aisle B occupancy clears below 70%.